

Bài toán data warehouse về bộ dữ liệu mẫu của Microsoft có CSDL:

<https://learn.microsoft.com/en-us/sql/samples/adventureworks-install-configure?view=sql-server-ver16&tabs=ssms>

https://dataedo.com/samples/html/AdventureWorks/doc/AdventureWorks_2/home.html

Sau khi tìm hiểu về bộ dữ liệu, có các yêu cầu như sau:

- Xây dựng 1 DW xoay quanh 6 câu hỏi xoay quanh Sale và Product Business Process như sau:
 - Report revenue by each employee => revenue
 - Report revenue by region => sales
 - Report number of orders by employee => Number of orders
 - Report number of units by region => Number of orders
 - Report sales quantity by product group => Number of products
 - Report sales quantity by product and region => Number of products
 - Tập trung vào 8 bảng của AW2012 database :
 - + Person
 - + Employee
 - + Territory
 - + Product Category
 - + Product Sub Category
 - + Product
 - + Sales Order Header
 - + Sales Order Detail
 - Ngoài ra có thể dùng thêm các bảng khác nếu cần thiết
- bài toán theo hình thái 1 và 2 của data warehouse, đi tìm dữ liệu để ETL, không phải là từ dữ liệu ra bài toán

Giải bài toán DW

Yêu cầu: Từ hệ thống bán hàng OLTP AdventureWorks → thiết kế 1 Data Warehouse để phân tích doanh thu, số lượng đơn hàng theo nhân viên, vùng và sản phẩm bằng Star Schema + Data Vault.

6 câu hỏi

- Report revenue by each employee → revenue, employee
- Report revenue by region → revenue, territory
- Report number of orders by employee → number of orders, employee
- Report number of units by region → number of orders, territory
- Report sales quantity by product group → number of orders, product group

- Report sales quantity by product and region → number of products, product, territory

→ Business process chính: Sales

Các chiều phân tích

- Employee
- Territory
- Product
- Date (ko yêu cầu nhưng bth DW sẽ có dim này, thông tin ở bảng Sales Order Header)

→ Fact table: Fact_Sales

Gồm các cột:

- SalesOrderID (từ SalesOrderHeader)
- ProductID (từ SalesOrderDetail)
- EmployeeID (=salespersonID từ SalesOrderHeader)
- TerritoryID (từ Sales.SalesTerritory được link với SalesOrderHeader)
- DateID (surrogate key từ OrderDate)
- OrderQty (từ SalesOrderDetail)
- UnitPrice (từ SalesOrderDetail)
- SalesAmount (doanh thu = revenue)

→ Các KPI cần tính

- + SalesAmount (=OrderQty×UnitPrice)
- + OrderQty (số lượng sản phẩm bán ra)
- + OrderCount (số lượng đơn hàng nhận về)

Grain statement (xác định 1 dòng trong fact table đại diện cho cái gì): 1 hàng trong fact_sales đại diện cho 1 sản phẩm trong 1 đơn hàng

→ **FactSales ETL**

INSERT INTO FactSales

SELECT

h.SalesOrderID,

d.ProductID,

h.SalesPersonID AS EmployeeID,

h.TerritoryID,

CONVERT(INT, FORMAT(h.OrderDate, 'yyyyMMdd')) AS DateID,

d.OrderQty,

```
d.UnitPrice,  
d.OrderQty * d.UnitPrice AS SalesAmount  
FROM Sales.SalesOrderHeader h  
JOIN Sales.SalesOrderDetail d  
ON h.SalesOrderID = d.SalesOrderID;
```

→ Dim tables

- DimEmployee
- DimTerritory
- DimProduct
- DimDate

1. DimEmployee

Từ bảng:

- HumanResources.Employee
- Person.Person
- Sales.SalesPerson

Column:

- EmployeeID
- EmployeeName
- JobTitle
- TerritoryID

→ SQL tạo DimEmployee

```
SELECT  
e.BusinessEntityID AS EmployeeID,  
p.FirstName + ' ' + p.LastName AS EmployeeName,  
e.JobTitle,  
s.TerritoryID  
FROM HumanResources.Employee e  
JOIN Person.Person p  
ON e.BusinessEntityID = p.BusinessEntityID  
LEFT JOIN Sales.SalesPerson s  
ON e.BusinessEntityID = s.BusinessEntityID;
```

2. DimProduct

Từ bảng:

- Production.Product
- Production.ProductSubcategory

- Production.ProductCategory

Column:

- ProductID
- ProductName
- ProductSubcategory
- ProductCategory

→ SQL tạo DimProduct

SELECT

```
p.ProductID,  
p.Name AS ProductName,  
ps.Name AS ProductSubcategory,  
pc.Name AS ProductCategory
```

FROM Production.Product p

LEFT JOIN Production.ProductSubcategory ps

ON p.ProductSubcategoryID = ps.ProductSubcategoryID

LEFT JOIN Production.ProductCategory pc

ON ps.ProductCategoryID = pc.ProductCategoryID;

3. DimTerritory

Từ bảng: Sales.SalesTerritory

Column:

- TerritoryID
- TerritoryName
- CountryRegionCode
- TerritoryGroup

→ SQL tạo DimTerritory

SELECT

```
TerritoryID,  
Name AS TerritoryName,  
CountryRegionCode,  
[Group] AS TerritoryGroup
```

FROM Sales.SalesTerritory;

4. DimDate

Tạo từ: SalesOrderHeader.OrderDate

Column:

- DateID
- Day

- Month
- Quarter
- Year

→ SQL tạo DimDate

SELECT DISTINCT

CONVERT(INT, FORMAT(OrderDate, 'yyyyMMdd')) AS DateID,

DAY(OrderDate) AS Day,

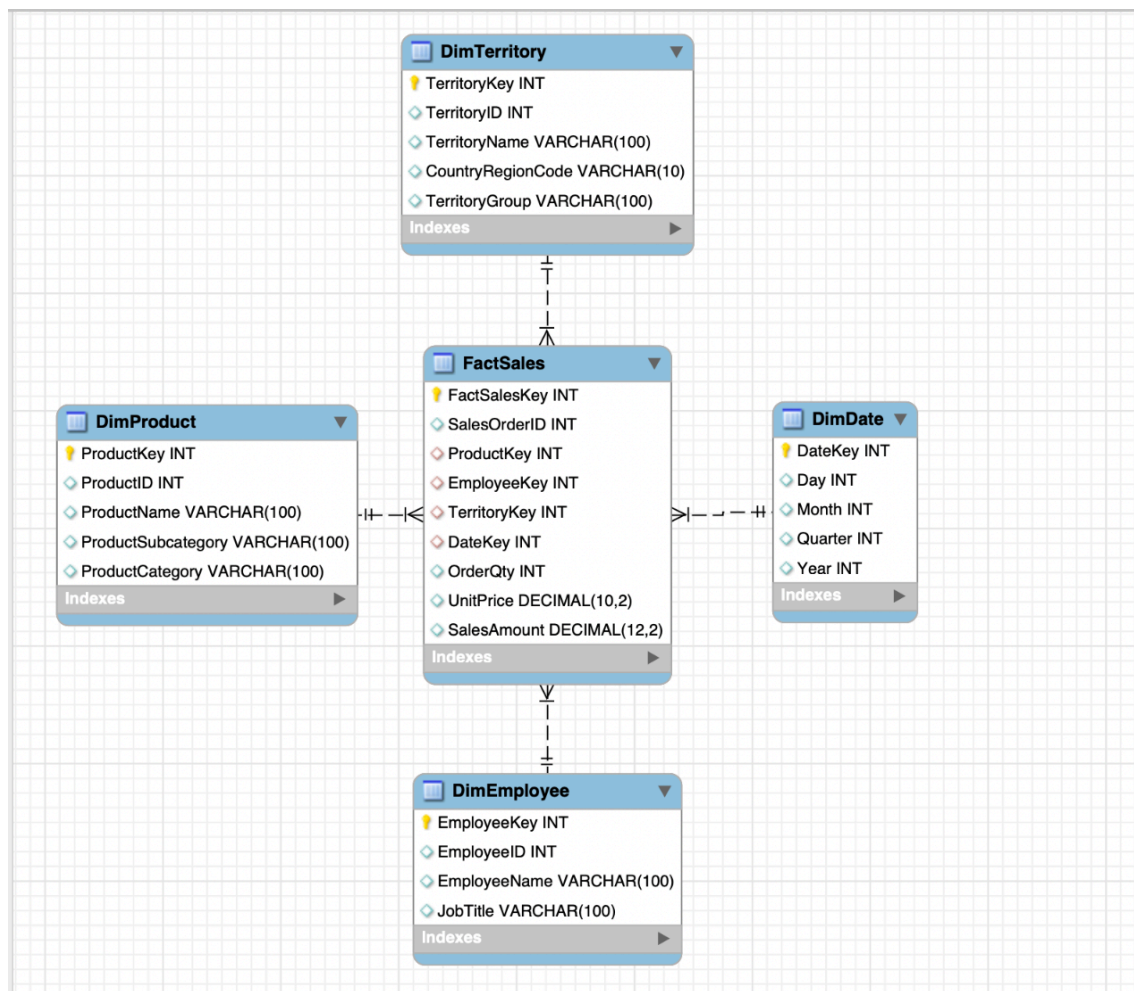
MONTH(OrderDate) AS Month,

DATEPART(QUARTER, OrderDate) AS Quarter,

YEAR(OrderDate) AS Year

FROM Sales.SalesOrderHeader;

Star Schema



Các yêu cầu:

1. Revenue by each employee

Sử dụng: FactSales + DimEmployee

→ SQL:

```
SELECT
    e.EmployeeName,
    SUM(f.SalesAmount) AS Revenue
FROM FactSales f
JOIN DimEmployee e
ON f.EmployeeID = e.EmployeeID
GROUP BY e.EmployeeName;
```

2. Revenue by region

```
SELECT
    t.TerritoryName,
    SUM(f.SalesAmount) AS Revenue
FROM FactSales f
JOIN DimTerritory t
ON f.TerritoryID = t.TerritoryID
GROUP BY t.TerritoryName;
```

3. Number of orders by employee

```
SELECT
    e.EmployeeName,
    COUNT(DISTINCT f.SalesOrderID) AS NumberOfOrders
FROM FactSales f
JOIN DimEmployee e
ON f.EmployeeID = e.EmployeeID
GROUP BY e.EmployeeName;
```

4. Number of units by region

```
SELECT
    t.TerritoryName,
    SUM(f.OrderQty) AS UnitsSold
FROM FactSales f
JOIN DimTerritory t
ON f.TerritoryID = t.TerritoryID
GROUP BY t.TerritoryName;
```

5. Sales quantity by product group

```
SELECT
    p.ProductCategory,
    SUM(f.OrderQty) AS QuantitySold
FROM FactSales f
JOIN DimProduct p
ON f.ProductID = p.ProductID
GROUP BY p.ProductCategory;
```

6. Sales quantity by product and region

```
SELECT
    p.ProductName,
    t.TerritoryName,
    SUM(f.OrderQty) AS QuantitySold
FROM FactSales f
JOIN DimProduct p
ON f.ProductID = p.ProductID
JOIN DimTerritory t
ON f.TerritoryID = t.TerritoryID
GROUP BY p.ProductName, t.TerritoryName;
```

Data Vault**a. Hubs**

- HubEmployee

→ Business key: BusinessEntityID

Column

- EmployeeHubKey
- BusinessEntityID
- LoadDate
- RecordSource

- HubProduct

→ Business key: ProductID

Column

- ProductHubKey
- ProductID
- LoadDate
- RecordSource

- HubTerritory

→ Business key: TerritoryID

Column

- TerritoryHubKey
- TerritoryID
- LoadDate
- RecordSource

- HubSalesOrder

Business key: SalesOrderID

Column

- SalesOrderHubKey
- SalesOrderID
- LoadDate

b. Links: Lưu mối quan hệ giữa các hubs

- LinkOrderProduct: Đơn hàng nào chứa sản phẩm nào.

- LinkOrderProductKey
- SalesOrderHubKey
- ProductHubKey
- LoadDate

- LinkOrderEmployee: Đơn hàng do nhân viên nào phụ trách.

- LinkOrderEmployeeKey
- SalesOrderHubKey
- EmployeeHubKey
- LoadDate

- LinkOrderTerritory: Đơn hàng thuộc territory nào.

- LinkOrderTerritoryKey
- SalesOrderHubKey
- TerritoryHubKey
- LoadDate

c. Satellites

- SatEmployee
- + EmployeeName
- + JobTitle
- + LoadDate

- SatProduct
 - + ProductName
 - + ProductCategory
 - + LoadDate
- SatSalesOrder
 - + OrderDate
 - + SalesAmount
 - + LoadDate
- SatTerritory
 - + TerritoryName
 - + CountryRegionCode
 - + LoadDate

Quá trình ETL

1. Extract:
 - Sales.SalesOrderHeader
 - Sales.SalesOrderDetail
 - Production.Product
 - HumanResources.Employee
 - Person.Person
 - Sales.SalesTerritory
2. Transform

Thực hiện join dữ liệu:

```
FROM Sales.SalesOrderHeader h
JOIN Sales.SalesOrderDetail d
ON h.SalesOrderID = d.SalesOrderID
```

```
JOIN Production.Product p
ON d.ProductID = p.ProductID
```

```
LEFT JOIN Sales.SalesPerson s
ON h.SalesPersonID = s.BusinessEntityID
```

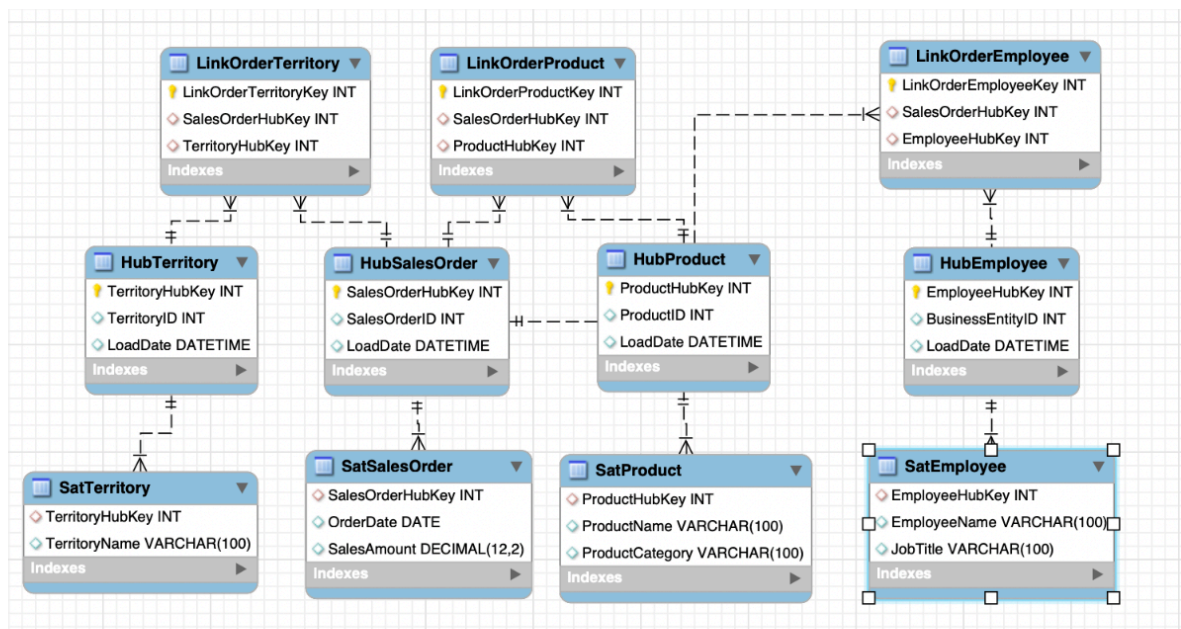
LEFT JOIN HumanResources.Employee e
ON s.BusinessEntityID = e.BusinessEntityID

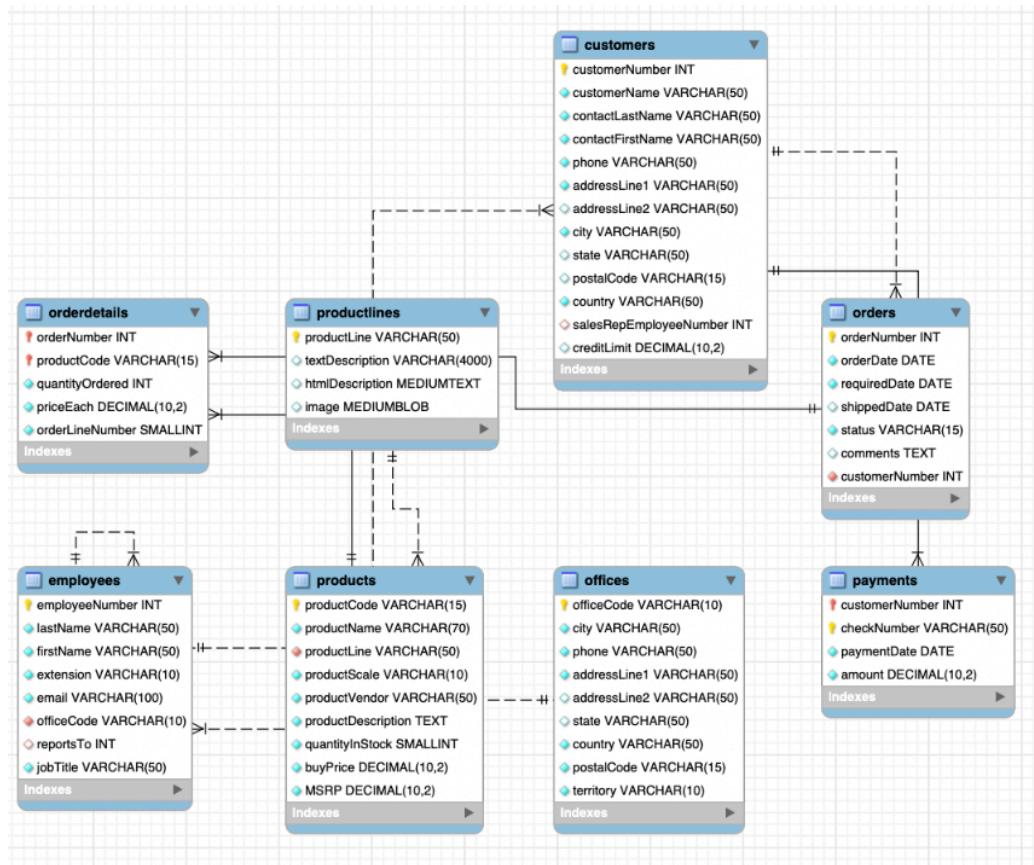
LEFT JOIN Sales.SalesTerritory t
ON h.TerritoryID = t.TerritoryID

3. Load

- FactSales
- DimEmployee
- DimProduct
- DimTerritory
- DimDate

Data Vault



CSDL Classicmodel:

Yêu cầu: Từ hệ thống bán hàng OLTP ClassicModels → thiết kế 1 Data Warehouse để phân tích doanh thu, đơn hàng và sản phẩm bằng Star Schema + Data Vault.

4 câu hỏi

- Report total revenue by customer
- Report number of orders by employee
- Report total quantity sold by product line
- Report revenue by office

→ Business process chính: Sales Process

Các chiều phân tích

- Customer
- Employee
- Product
- Office
- Date

→ Fact table: FactSales

Gồm các cột

- OrderNumber
- ProductCode
- CustomerNumber
- EmployeeNumber
- OfficeCode
- DateID
- QuantityOrdered
- PriceEach
- SalesAmount

Lấy thông tin từ

- orders
- orderdetails
- customers
- employees
- offices
- products

→ Dim tables

- DimCustomer
- DimEmployee
- DimProduct
- DimOffice
- DimDate

1. DimCustomer

Từ bảng: customers

Column

- CustomerNumber
- CustomerName
- Country
- City
- CreditLimit

```
SELECT  
customerNumber,  
customerName,  
country,  
city,  
creditLimit  
FROM customers;
```

2. DimEmployee

Từ bảng

- employees
- offices

Column

- EmployeeNumber
- EmployeeName
- JobTitle
- OfficeCode
- OfficeCity

```
SELECT  
e.employeeNumber,  
CONCAT(e.firstName, ' ', e.lastName) AS EmployeeName,  
e.jobTitle,  
o.officeCode,  
o.city AS OfficeCity  
FROM employees e  
JOIN offices o  
ON e.officeCode = o.officeCode;
```

3. DimProduct

Từ bảng: products

Column

- ProductCode
- ProductName

- ProductLine
- ProductVendor

```
SELECT  
productCode,  
productName,  
productLine,  
productVendor  
FROM products;
```

4. DimOffice

Từ bảng: offices

Column

- OfficeCode
- City
- Country
- Territory

```
SELECT  
officeCode,  
city,  
country,  
territory  
FROM offices;
```

5. DimDate

Tạo từ: orders.orderDate

```
SELECT DISTINCT  
DATE_FORMAT(orderDate, '%Y%m%d') AS DateID,  
orderDate AS FullDate,  
MONTH(orderDate) AS Month,  
QUARTER(orderDate) AS Quarter,  
YEAR(orderDate) AS Year  
FROM orders;
```

Grain statement (xác định 1 dòng trong fact table đại diện cho cái gì): 1 hàng trong fact_sales đại diện cho 1 sản phẩm trong 1 đơn hàng

→ FactSales ETL

```
INSERT INTO FactSales
```

```
SELECT
```

```
o.orderNumber,
```

```
od.productCode,
```

```
c.customerNumber,
```

```
e.employeeNumber,
```

```
o2.officeCode,
```

```
DATE_FORMAT(o.orderDate, '%Y%m%d') AS DateID,
```

```
od.quantityOrdered,
```

```
od.priceEach,
```

```
od.quantityOrdered * od.priceEach AS SalesAmount
```

```
FROM orders o
```

```
JOIN orderdetails od
```

```
ON o.orderNumber = od.orderNumber
```

```
JOIN customers c
```

```
ON o.customerNumber = c.customerNumber
```

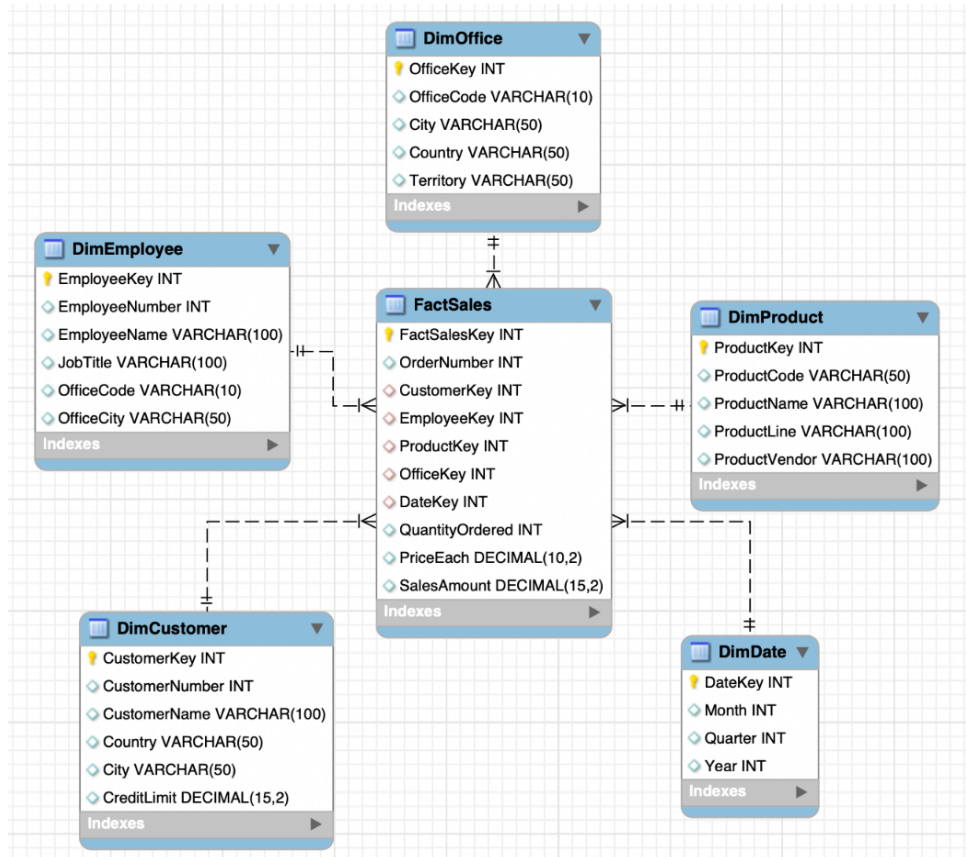
```
JOIN employees e
```

```
ON c.salesRepEmployeeNumber = e.employeeNumber
```

```
JOIN offices o2
```

```
ON e.officeCode = o2.officeCode;
```

Star Schema



→ Các yêu cầu

1. Revenue by customer

```

SELECT
    c.CustomerName,
    SUM(f.SalesAmount) AS Revenue
FROM FactSales f
JOIN DimCustomer c
ON f.CustomerNumber = c.CustomerNumber
GROUP BY c.CustomerName;
  
```

2. Number of orders by employee

```

SELECT
    e.EmployeeName,
    COUNT(DISTINCT f.OrderNumber) AS NumberOfOrders
FROM FactSales f
JOIN DimEmployee e
ON f.EmployeeNumber = e.EmployeeNumber
GROUP BY e.EmployeeName;
  
```


3. Total quantity sold by product line

```
SELECT
  p.ProductLine,
  SUM(f.QuantityOrdered) AS QuantitySold
FROM FactSales f
JOIN DimProduct p
ON f.ProductCode = p.ProductCode
GROUP BY p.ProductLine;
```

4. Revenue by office

```
SELECT
  o.City,
  SUM(f.SalesAmount) AS Revenue
FROM FactSales f
JOIN DimOffice o
ON f.OfficeCode = o.OfficeCode
GROUP BY o.City;
```

Data Vault

- a. Hubs
 - HubCustomer
 - HubEmployee
 - HubProduct
 - HubOrder
- b. Links
 - LinkOrderCustomer
 - LinkOrderProduct
 - LinkOrderEmployee
- c. Satellites

SatCustomer

- CustomerName
- Country
- CreditLimit
- LoadDate

SatEmployee

- EmployeeName

- JobTitle
- OfficeCode
- LoadDate

SatProduct

- ProductName
- ProductLine
- ProductVendor
- LoadDate

SatOrder

- OrderDate
- SalesAmount
- QuantityOrdered
- LoadDate

→ Data Vault:

